

1. Executive Overview & Mission

VendLex Kenya (<https://vendlex.co.ke>) is a unified digital commerce engine engineered specifically for the Kenyan economy. The platform bridges the trust, financial, and logistical gaps between local businesses, certified service tradespeople, and millions of consumers across all **47 Kenyan Counties**. By merging verified multi-vendor marketplace retail with on-demand technician dispatch and an integrated merchant SaaS management operating system, VendLex serves as a full-stack commercial operating system for Kenya.

2. The Three Strategic Pillars: SHOP • GROW • PROSPER

Pillar	Target Audience	Core Deliverables & Capabilities	Key Innovations
01. SHOP <i>Consumer Retail</i>	Kenyan Shoppers & Corporate Buyers	Multi-category shopping: Electronics, African Fashion & Kitenge, Solar PV Gear, Home appliances, and agricultural supplies with instant Lipa na M-Pesa STK push.	Direct WhatsApp merchant contact, same-day delivery dispatch, fraud reporting, and wishlist sync.
02. GROW <i>Merchant SaaS</i>	Retailers, Wholesalers, Artisans & SMEs	5-minute online store setup, automated KRA eTIMS compliant PDF invoicing, multi-channel order tracking, inventory alerts, and sales analytics.	Bilingual Swahili/English AI Business Assistant for automated social ads and customer chat scripts.
03. PROSPER <i>Service Network</i>	Plumbers, Electricians, Techs & Clients	On-demand dispatch of verified local tradespeople, fixed starting quote previews, certified skills credentials, and direct booking workflow.	Ratings, verified photo reviews, and county-level GPS technician discovery.

3. Financial Infrastructure & Safaricom Daraja 2.0 Integration

VendLex has a native, production-tested integration with Safaricom Daraja 2.0 APIs. The platform natively supports **Buy Goods Till Numbers, Paybills, and Pochi la Biashara** accounts for real-time customer checkout, merchant onboarding fee collection, and escrow disbursements.

API Endpoint	Module & Path	Operational Description
STK Push (Lipa na M-Pesa)	/api/mpesa/stkpush lib/mpesa/daraja.ts	Triggers SIM push notification to customer phone with 45-second countdown. Validates Kenyan phone numbers (254...) and bounds with sliding-window Edge rate-limiting.
Payment Callback Webhook	/api/mpesa/callback	Asynchronously receives Safaricom transaction confirmation receipts (e.g., SKLQG89124), updates store activation status, and logs receipts to the platform ledger.
Transaction Query API	/api/mpesa/query	Handles automated polling to resolve edge-case network dropouts or delayed customer PIN entries.
Escrow & Transaction Ledger	lib/mpesa/transaction-store.ts	Tracks transaction volume, subscription payments, and customer receipts across all merchants.

4. Defensive Security Architecture & Edge Middleware

VendLex employs defense-in-depth security engineered for e-commerce and high-throughput financial transactions:

- **Edge Sliding-Window Rate Limiting (middleware.ts):** Protects payment endpoints against credential stuffing and SIM bombing (15 req/min on STK push, 60 req/min on standard endpoints). Returns standard HTTP 429 with Retry-After headers.
- **Strict Security Headers (next.config.ts):** Implements HTTP Strict Transport Security (HSTS max-age=63072000; includeSubDomains; preload), X-Frame-Options: DENY, X-Content-Type-Options: nosniff, and rigorous Content Security Policy (CSP).
- **Kenyan Regulatory & PII Masking (lib/security.ts):** Strict regex verification for Kenyan numbers (^(?:254|\\+254|0)?([17]\\d{8})\$) with PII phone masking (e.g. 0712***678) in customer logs.
- **Timing-Safe Authentications:** Constant-time comparisons (crypto.timingSafeEqual) to prevent side-channel timing attacks on API keys and passwords.

5. Full Technology Stack & Build Performance

Layer	Technology Selected	Architectural Benefit
Core Web Framework	Next.js 15.1 (App Router), React 19, TypeScript 5	Zero-runtime client JS overhead for static components; fast server-side streaming for dynamic routes.
Design System & UI	Tailwind CSS 3.4 + Fluid Clamp Typography CSS	WCAG AAA contrast tokens, responsive fluid typography clamp scales, glassmorphism, and dark/light mode persistence.
Micro-Interactions	CSS3 Keyframes + Framer Motion 12	Smooth card elevation, wishlist heart pop, automatic ambient slideshow, and reduced-motion accessibility.
Mobile App Distribution	PWA Web Manifest + Trusted Web Activity (TWA) / Capacitor	Google Play Store distributable without maintaining separate native Android codebases.
Production Build	Clean 100% compilation across 37 routes	Lightweight 6.41 kB homepage first-load JS; 0 TypeScript errors or lint warnings.

6. Platform Directory & Key Route Manifest

Route Path	Module Description & Capabilities	Access Level
/	Streamlined Homepage: Automatic ambient commerce slideshow, 3 Pillars (SHOP • GROW • PROSPER), 47-county category visual grid, trending deals, verified stores, service showcase, and social proof.	Public
/marketplace	Full catalog discovery with instant price filtering, county filters, ratings, and instant cart dispatch.	Public
/businesses	Directory of verified Kenyan stores, manufacturers, and artisans across all 47 counties.	Public
/services	Certified Kenyan service technicians (electricians, plumbers, solar engineers) with quote requests.	Public
/pricing	Transparent merchant subscription tiers: Free, Starter (KSh 299), Business (KSh 799), Pro (KSh 1,499).	Public
/seller/onboarding	5-step merchant onboarding with live M-Pesa STK push fee activation and KYC verification upload.	Merchant
/seller/dashboard	Merchant operating center: live inventory, KRA eTIMS invoice generation, order management, and AI marketing tools.	Merchant
/admin	Platform administration: seller KYC review, escrow transaction reconciliation, and fraud monitoring.	Admin

Platform: VendLex Kenya • **Tagline:** SHOP • GROW • PROSPER

Domain: <https://vendlex.co.ke> • **Primary Currency:** Kenyan Shilling (KES / KSh)

Compliance: KRA eTIMS, Data Protection Act 2019, Safaricom Daraja 2.0 OpenAPI Specification

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